Coronary Heart Disease Risk Factors: Framingham Heart Study Analysis

Summary

Of 4,434 Framingham Heart Study participants, 1,240 (28%) developed coronary heart disease (CHD). The clearest differences were age, sex and blood pressure: CHD rose from 16.3% at ages 30–39 to 42.3% at 60 and over, men had a much higher rate than women (36.5% vs 21.3%), and 45.3% of the CHD group had hypertension compared with 27.2% of the No CHD group.

Diabetes was about three times as common in the CHD group (5.4% vs 1.7%), and BMI was slightly higher (average 27 vs 25). Smoking status alone showed almost no difference (28.3% vs 27.7%). All correlations with CHD were weak (below 0.25), so these are associations, not causes.

1. Introduction

Coronary heart disease is a leading cause of illness and death, and the Framingham Heart Study is one of the best-known studies of its risk factors. This report explores a Framingham dataset to answer one question: **which participant characteristics are associated with developing CHD, and how large are the differences between people who did and did not develop it?**

The analysis is exploratory. It describes patterns in the data and sets up questions for further modelling.

2. Data

The dataset (`framingham_2024-1.csv`) has 4,434 participants and 26 fields. It was provided for coursework, so the raw data is not shared with this report.

Group	Variables analysed
Outcome	Any CHD (1 = developed CHD, 0 = did not)
Demographics	Age, sex

Group	Variables analysed
Lifestyle	Current smoker, cigarettes per day
Clinical measurements	Systolic BP, diastolic BP, BMI, heart rate, total cholesterol, glucose
Medical history	Diabetes, hypertension, prior stroke, on BP medication

These 15 of the 26 fields were analysed. 19 participants have no BMI recorded and are left out of the BMI analysis. Other missing values are left out of the measures they affect.

3. Method

1. **Group comparison:** participants were split into No CHD (3,194) and CHD (1,240), and each variable was compared between the two groups.
2. **Averages and rates:** continuous measures (age, BMI, blood pressure, heart rate, cigarettes per day) use group averages. Yes/no fields use the share of participants with that condition, the average of a 0/1 field.
3. **Age groups:** 30–39, 40–49, 50–59 and 60+. Ages 70–79 were merged into 60+ because only 2 participants fell in that range, too few for a reliable rate.
4. **Correlation analysis:** a correlation matrix of the 15 variables measured how strongly each one moves with CHD.
5. **Visualisation:** charts and an interactive dashboard were built in Tableau Desktop, using calculated fields for the age groups and rates.

4. Findings

4.1 Profile of the two groups

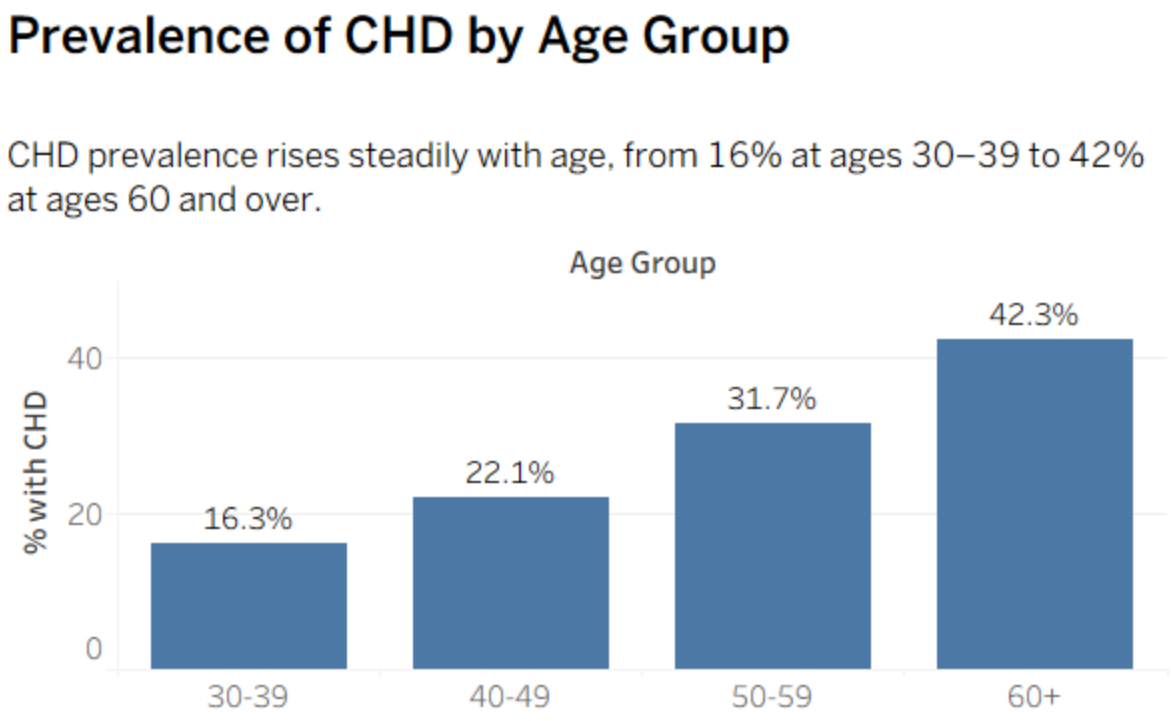
Participants with CHD were older, had higher blood pressure, and more often had hypertension, diabetes or a prior stroke. Heart rate and cigarettes per day barely differed.

Indicator	No CHD	CHD
Number of people	3,194	1,240
Average age	49	53

Indicator	No CHD	CHD
Average BMI	25	27
Average systolic BP (mmHg)	130	140
Average diastolic BP (mmHg)	82	86
Average heart rate (bpm)	76	76
Average cigarettes per day	9	10
On BP medication	2.2%	6.0%
Diabetes	1.7%	5.4%
Prior stroke	0.5%	1.3%
Hypertension	27.2%	45.3%

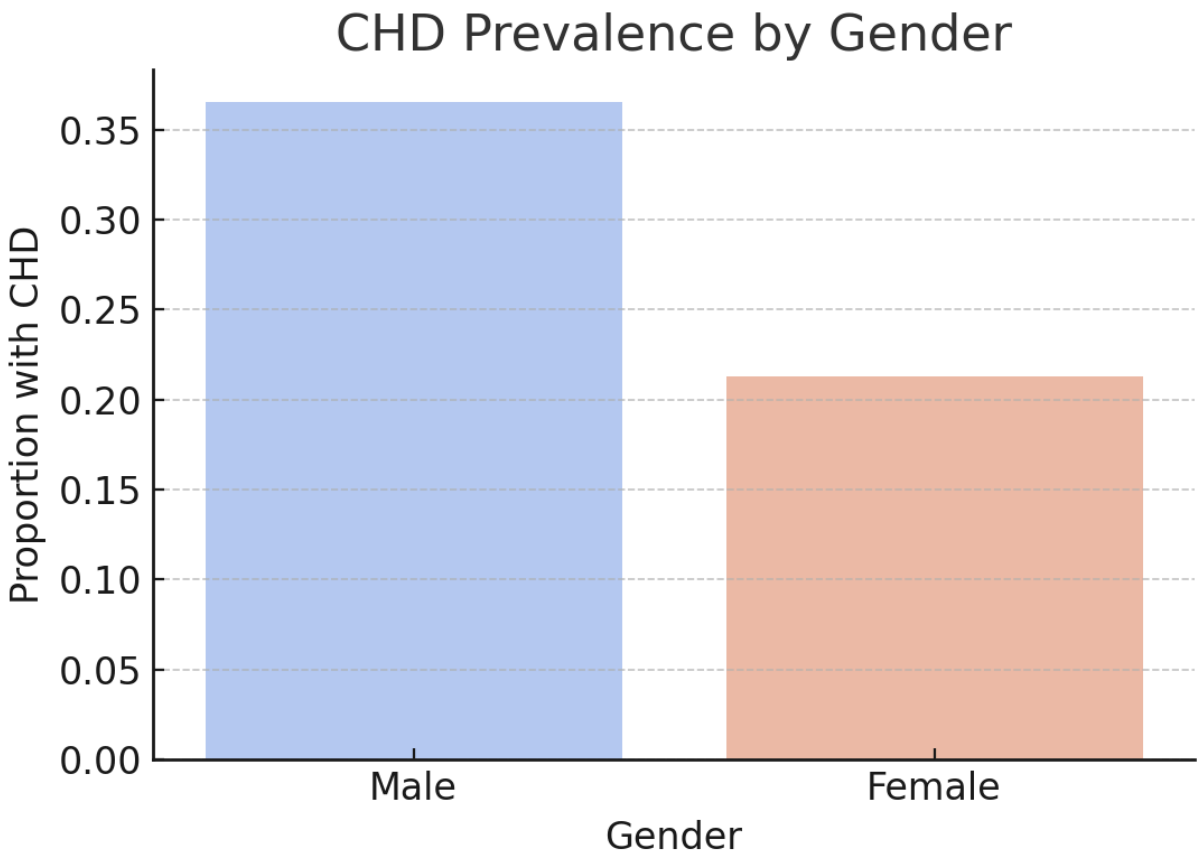
4.2 Age

CHD prevalence rises steadily with age, from 16.3% at ages 30–39 to 42.3% at 60 and over, about 2.6 times as high.



4.3 Sex

Men had a much higher CHD rate than women, 36.5% compared with 21.3%. This is the largest difference between any two groups in the analysis.

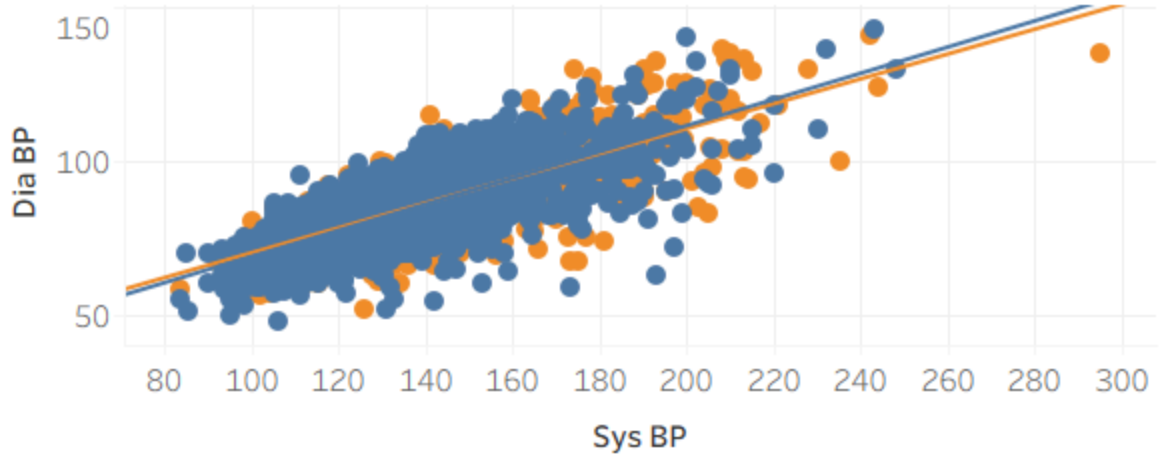


4.4 Blood pressure

Systolic and diastolic blood pressure rise together in both groups. The CHD group sits higher on average (systolic 140 vs 130), and nearly half of it had hypertension.

Systolic vs Diastolic Blood Pressure by CHD status

Each dot is a participant (blue = No CHD, orange = CHD). Both blood pressure readings rise together in both groups. On average, people with CHD had higher systolic blood pressure (140 vs 130).

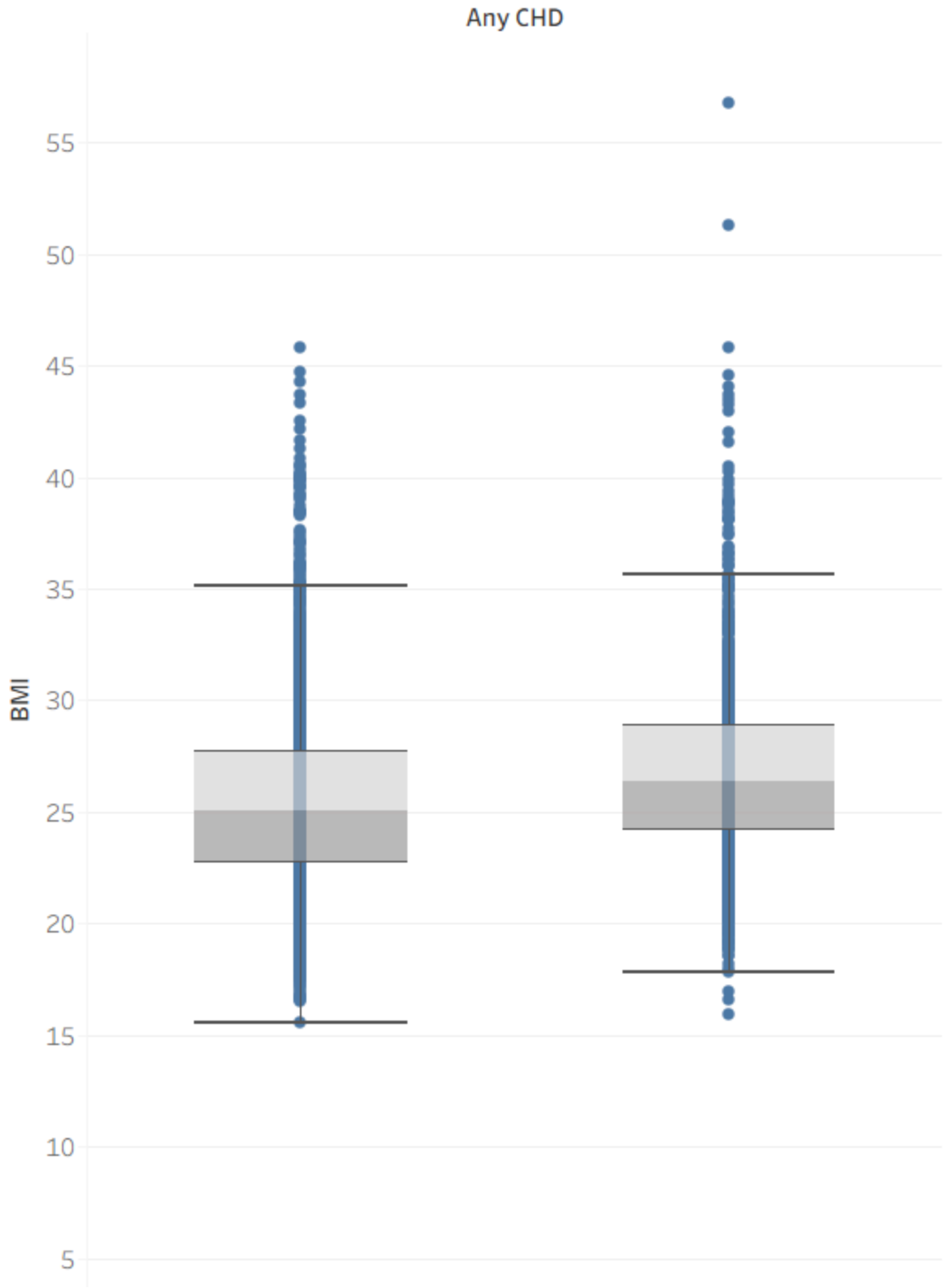


4.5 BMI

The CHD group's BMI distribution is shifted slightly upward (average 27 vs 25), but the two groups overlap heavily.

BMI vs. CHD

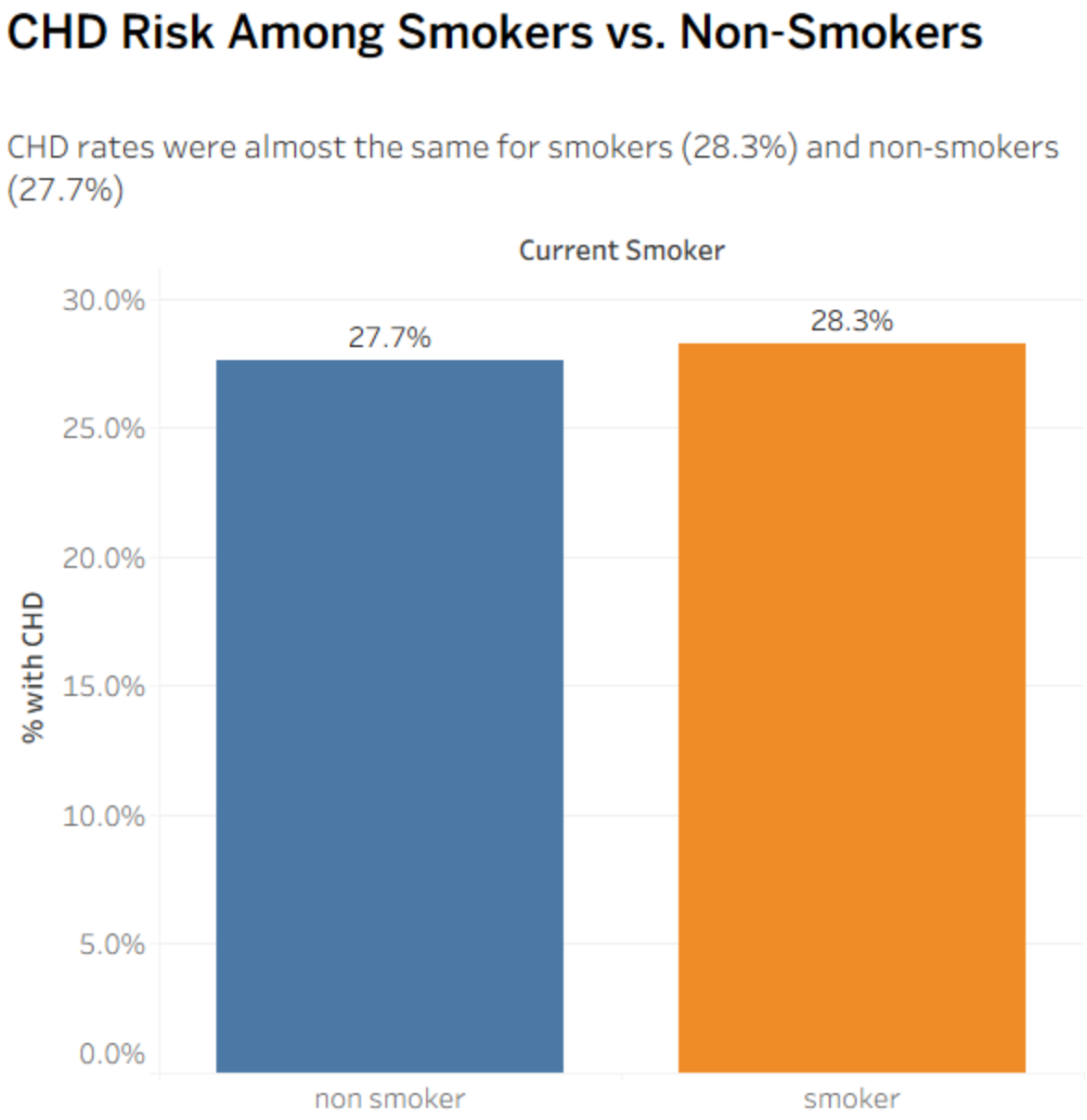
Box plots show the spread of BMI in each group. People with CHD had a slightly higher BMI on average (27 vs 25).





4.6 Smoking

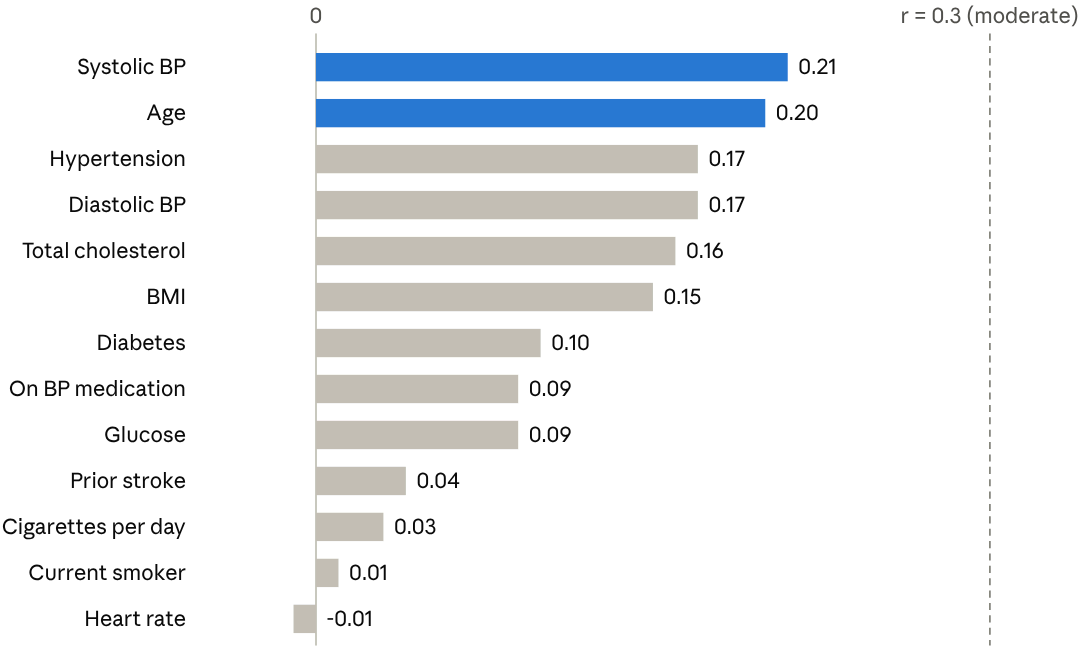
CHD rates were almost the same for smokers (28.3%) and non-smokers (27.7%). One possible reason is that smoking differs by age; this is tested in the next steps rather than assumed.



5. Correlations with CHD

Systolic BP and age had the strongest links to CHD, but all were weak

Correlation (r) of each variable with Any CHD



Any CHD row of the original analysis's correlation matrix · 13 variables (sex shown in 4.3)

Every correlation is below 0.3, the level usually treated as moderate, so no single variable separates the groups well on its own. Smoking status and cigarettes per day show almost no correlation, matching the near-identical rates in 4.6.

6. Dashboard

The Tableau dashboard brings the main views together in one place: the risk scorecard, CHD by age group, blood pressure, BMI and smoking.

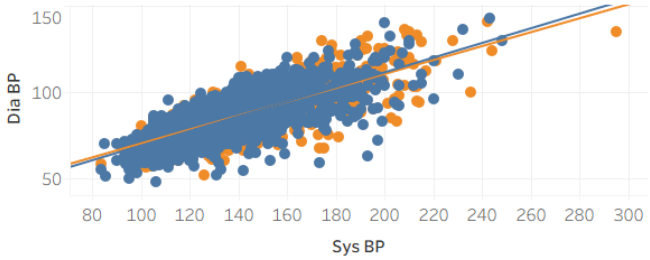
CHD Risk Scorecard

Key indicators of CHD risk in the population

	Any CHD	
	NO CHD	CHD
Number Of People	3,194	1,240
Avg. Age	49	53
Avg. BMI	25	27
Avg. Sys BP	130	140
Avg. Dia BP	82	86
Avg. Heart Rate	76	76
Avg. Cigs Per Day	9	10
On BP Medication(%)	2.2%	6.0%
Diabetes(%)	1.7%	5.4%
Prior Stroke(%)	0.5%	1.3%
Hypertension(%)	27.2%	45.3%

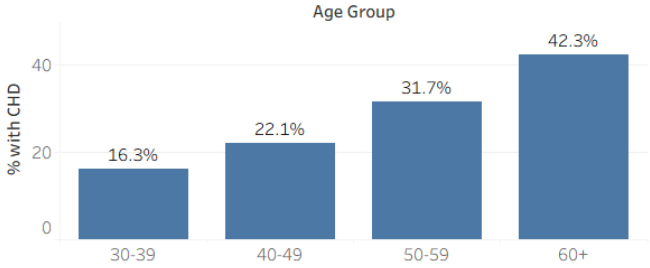
Systolic vs Diastolic Blood Pressure by CHD status

Each dot is a participant (blue = No CHD, orange = CHD). Both blood pressure readings rise together in both groups. On average, people with CHD had higher systolic blood pressure (140 vs 130).



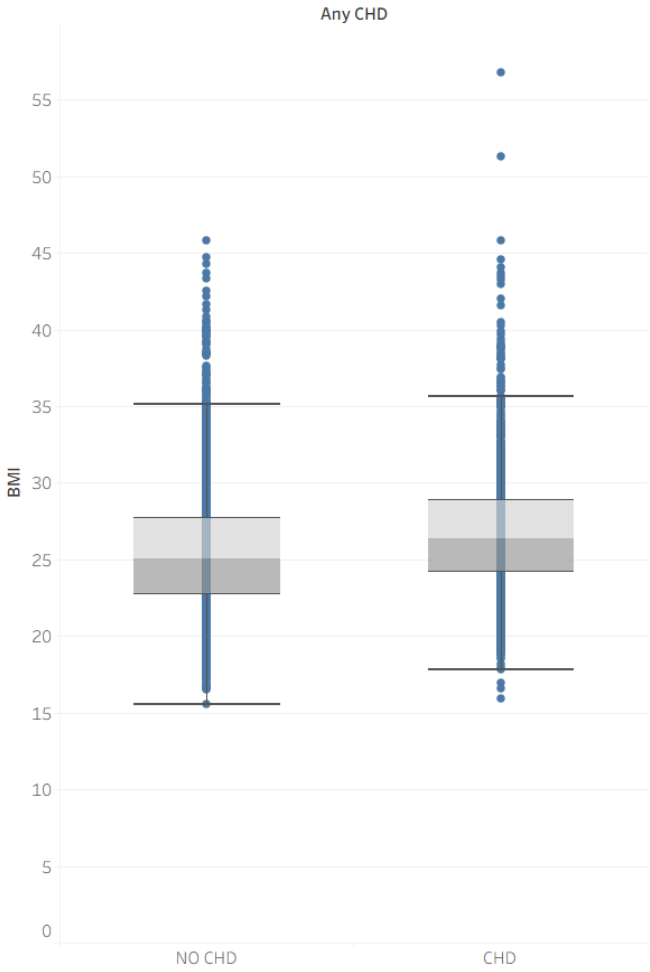
Prevalence of CHD by Age Group

CHD prevalence rises steadily with age, from 16% at ages 30–39 to 42% at ages 60 and over.



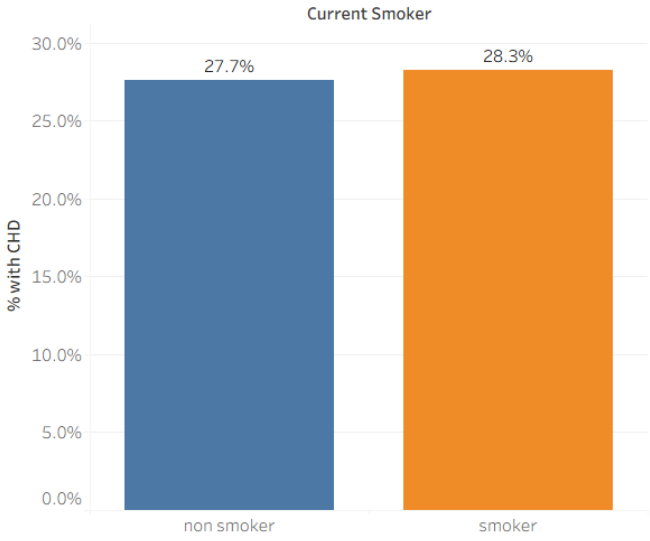
BMI vs. CHD

Box plots show the spread of BMI in each group. People with CHD had a slightly higher BMI on average (27 vs 25).



CHD Risk Among Smokers vs. Non-Smokers

CHD rates were almost the same for smokers (28.3%) and non-smokers (27.7%)



7. Limitations and next steps

Limitations

- The results show associations only. No statistical tests or models were applied, so no difference here is shown to be statistically significant or causal.
- Each factor was looked at on its own. Factors overlap (for example, older participants also tend to have higher blood pressure), so one factor can hide or exaggerate another.
- Participants with missing values are left out of the measures they affect.

Next steps

1. Compare CHD rates for smokers and non-smokers within each age group, to check whether age hides an effect of smoking.
2. Fit a logistic regression with age, sex, systolic BP, diabetes and cholesterol together, to estimate each factor's effect while holding the others constant.
3. Test the main group differences for statistical significance.

8. Conclusion

Age, sex and blood pressure show the clearest links to CHD in this dataset. CHD prevalence more than doubled from the youngest to the oldest age group, men had a much higher rate than women, and hypertension was far more common among those with CHD. Diabetes and BMI showed smaller differences, and smoking status alone showed almost none.

Because every correlation is weak and the factors overlap, the next stage is a model that weighs them together. That would show which factors matter most once the others are taken into account, and help target prevention at the highest-risk groups.